Project Proposal

Traffic Sign Detection Using Deep Learning for Intelligent Transport Systems

Group G17

This project aims to develop an accurate traffic sign detection system using modern computer vision techniques. We will compare a classical image processing approach with a deep learning object detection model to evaluate their performance.

Problem Statement

Develop a computer vision system that automatically detects traffic signs from road images by identifying their locations using bounding boxes. The system should perform accurately under different road conditions and be evaluated using standard object detection metrics.

Main Objective

Develop an efficient traffic sign detection system using deep learning and compare it with a traditional computer vision approach.

Motivation

Traffic sign detection plays a significant role in intelligent transportation systems. A reliable detection system can improve driver safety, assist autonomous vehicles, and reduce traffic accidents. This project also provides practical experience with object detection, dataset preparation, deep learning model training and performance evaluation.

Related Work

Previous studies have successfully applied deep learning models such as YOLO, SSD, and Faster R-CNN for traffic sign detection. Traditional methods based on color thresholding and shape detection have also been used but generally perform poorly under complex lighting and background conditions. Recent YOLO models provide high detection accuracy while maintaining real-time performance, making them suitable for this project.

Dataset

German Traffic Sign Detection Benchmark (GTSDB). The official train/test split will be used.

Methodology

The project will use the GTSDB dataset for traffic sign detection. First, the images will be preprocessed and augmented to improve model performance. A classical computer vision approach using HSV color thresholding and contour detection will be implemented as the baseline. Then, a pretrained YOLOv8 model will be fine-tuned using transfer learning. Finally, the performance of both methods will be compared.

Evaluation Plan

The proposed system will be evaluated using mAP@0.5, Precision and Recall on the test dataset. Qualitative results will also be presented by comparing detected traffic signs with the ground truth. The performance of the fine-tuned YOLOv8 model will be compared against the classical baseline to measure the improvement.

Risks and Mitigation

Risk	Mitigation
Limited GPU resources	Use Google Colab and YOLOv8 Nano model
Small dataset	Apply data augmentation and transfer learning
Overfitting	Early stopping and validation monitoring
Annotation errors	Verify dataset annotations before training
Time constraints	Follow the planned weekly schedule

Timeline

Week	Activity	Deliverable
Week 1 (8–14 Jul)	Literature review, dataset selection, GitHub repository setup, project proposal	Project proposal
Week 2 (15–21 Jul)	Download and preprocess the GTSDB dataset, perform exploratory data analysis	Prepared dataset
Week 3 (22–28 Jul)	Implement the classical computer vision baseline and pretrained YOLO baseline	Baseline results
Week 4 (29 Jul–4 Aug)	Fine-tune the YOLOv8 model	Initial trained model
Week 5 (5–11 Aug)	Improve the model using data augmentation and parameter tuning	Improved model
Week 6 (12–18 Aug)	Evaluate the model and compare it with the baseline	Performance evaluation
Week 7 (19 Aug–1 Sep)	Prepare the report, presentation, and demonstration	Draft report and presentation
Week 8 (2–7 Sep)	Final testing, documentation, and project submission	Final report, GitHub repository, and presentation